Exp 39-Display the First 3 Rows of a DataFrame Using Pandas

Aim

To write a Pandas program to create a DataFrame from given dictionary data and display its first 3 rows.

Procedure

1. Import the Pandas library.
2. Create a dictionary containing names, scores, attempts, and qualification status.
3. Create a DataFrame using `pd.DataFrame()`.
4. Use the `head(3)` function to retrieve the first 3 rows.
5. Display the result.

Code

39.py - C:/Users/subha/OneDrive/Desktop/Query processing/39.py (3.12.10)

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```python
import pandas as pd
import numpy as np

exam_data = {
    'name': ['Anastasia', 'Dima', 'Katherine', 'James', 'Emily',
             'Michael', 'Matthew', 'Laura', 'Kevin', 'Jonas'],
    'score': [12.5, 9, 16.5, np.nan, 9, 20, 14.5, np.nan, 8, 19],
    'attempts': [1, 3, 2, 3, 2, 3, 1, 1, 2, 1],
    'qualify': ['yes', 'no', 'yes', 'no', 'no',
                'yes', 'yes', 'no', 'no', 'yes']
}

labels = ['a', 'b', 'c', 'd', 'e', 'f', 'g', 'h', 'i', 'j']

df = pd.DataFrame(exam_data, index=labels)

print("First 3 rows of the DataFrame:")
print(df.head(3))
```

Output

```
First 3 rows of the DataFrame:
        name  score  attempts qualify
a  Anastasia   12.5         1     yes
b       Dima    9.0         3      no
c  Katherine   16.5         2     yes
```

Result

Thus, the Pandas program was successfully executed, and the first 3 rows of the DataFrame were displayed using the `head(3)` function.

Exp 40-Select the 'name' and 'score' Columns from a DataFrame

Aim

To write a Pandas program to select the 'name' and 'score' columns from a given DataFrame.

Procedure

1. Import the Pandas and NumPy libraries.
2. Create the given dictionary data.
3. Create a DataFrame using `pd.DataFrame()`.
4. Select the `name` and `score` columns using `df[['name', 'score']]`.
5. Display the selected columns.

Code

40.py - C:/Users/subha/OneDrive/Desktop/Query processing/40.py (3.12.10)

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```
import pandas as pd
import numpy as np

exam_data = {
    'name': ['Anastasia', 'Dima', 'Katherine', 'James', 'Emily',
             'Michael', 'Matthew', 'Laura', 'Kevin', 'Jonas'],
    'score': [12.5, 9, 16.5, np.nan, 9, 20, 14.5, np.nan, 8, 19],
    'attempts': [1, 3, 2, 3, 2, 3, 1, 1, 2, 1],
    'qualify': ['yes', 'no', 'yes', 'no', 'no',
                'yes', 'yes', 'no', 'no', 'yes']
}

labels = ['a', 'b', 'c', 'd', 'e', 'f', 'g', 'h', 'i', 'j']

df = pd.DataFrame(exam_data, index=labels)

print("Selected name and score columns:")
print(df[['name', 'score']])
```

Output

```
Selected name and score columns:
        name  score
a  Anastasia   12.5
b       Dima    9.0
c  Katherine   16.5
d      James    NaN
e      Emily    9.0
f    Michael   20.0
g    Matthew   14.5
h      Laura    NaN
i      Kevin    8.0
j      Jonas   19.0
```

Result

Thus, the 'name' and 'score' columns were successfully selected and displayed from the given DataFrame.